

Machine Learning, Deep Learning & Agentic AI

Study notes | Page 1 of 2

1. Machine Learning (ML)

Definition: ML is a branch of AI where computers learn patterns from data to make predictions or decisions, instead of being explicitly programmed with rules. A model is trained on examples, then applied to new, unseen data.

Main types of learning

Type	Idea	Examples
Supervised	Learn from labeled data (input + correct answer).	Spam detection, house-price prediction, image classification
Unsupervised	Find hidden structure in unlabeled data.	Customer clustering (K-Means), PCA, anomaly detection
Reinforcement	An agent learns by trial and error using rewards and penalties.	Game-playing AI, robotics, recommendation tuning
Self-supervised	Labels are created from the data itself.	Predicting the next word (LLM pre-training)

Typical ML workflow

- **Collect and clean data** - handle missing values, duplicates and outliers.
- **Feature engineering** - select or create useful input variables; scale/normalize.
- **Split data** - training / validation / test sets (e.g. 70/15/15).
- **Train and tune** - fit the model, tune hyperparameters (learning rate, tree depth).
- **Evaluate and deploy** - measure on the test set, then monitor for drift in production.

Key concepts

- **Common algorithms:** Linear/Logistic Regression, k-NN, Decision Trees, Random Forest, SVM, Gradient Boosting (XGBoost).
- **Overfitting:** model memorizes training data and fails on new data. **Underfitting:** model is too simple to capture the pattern. Fix with more data, regularization, cross-validation, simpler/complex models.
- **Bias-variance trade-off:** high bias = underfit; high variance = overfit; aim for the balance.
- **Metrics:** Accuracy, Precision, Recall, F1-score, ROC-AUC (classification); MAE, MSE, R^2 (regression).
- **Loss + optimization:** the loss measures error; gradient descent updates parameters to minimize it.

2. Deep Learning (DL)

Definition: DL is a subset of ML that uses **artificial neural networks with many layers** to automatically learn features from raw data (images, text, audio). It excels when data and compute are plentiful.

How a neural network learns

- **Neuron:** computes output = activation(weights x inputs + bias). Layers = input, hidden, output.
- **Activation functions:** ReLU, Sigmoid, Tanh, Softmax - add non-linearity so networks can model complex patterns.
- **Forward pass** produces a prediction; **loss function** (cross-entropy, MSE) measures the error.
- **Backpropagation** computes gradients; an **optimizer** (SGD, Adam) updates weights. One pass over all data = 1 epoch.
- **Regularization:** Dropout, weight decay, early stopping, batch normalization, data augmentation.

Major architectures

Architecture	Best for	Key idea
CNN	Images, video, sign recognition	Convolution filters detect local patterns (edges, shapes).
RNN / LSTM / GRU	Sequences, time series	Keep a hidden state (memory) across time steps.
Transformer	Language, vision, multimodal	Self-attention lets every token relate to every other token in parallel.
GAN / Diffusion	Image and media generation	Learn to generate new, realistic data.
Autoencoder	Compression, denoising	Encode data to a small representation, then reconstruct.

Tools: Python, PyTorch, TensorFlow/Keras, scikit-learn, NumPy, Pandas; GPUs/TPUs for training. **Transfer learning** (fine-tuning a pre-trained model) saves time and data.

Machine Learning, Deep Learning & Agentic AI

Study notes | Page 2 of 2

3. Large Language Models (the foundation of Agentic AI)

LLMs are Transformer-based deep learning models trained on huge text corpora to predict the next token. After pre-training they are **fine-tuned** and aligned (instruction tuning, RLHF) to follow instructions. Useful techniques: **prompt engineering**, **RAG** (Retrieval-Augmented Generation: fetch relevant documents and give them to the model to reduce hallucination), and **fine-tuning** (e.g. LoRA) for specific domains or languages such as Kiswahili.

4. Agentic AI

Definition: Agentic AI refers to systems that **pursue a goal autonomously** - they plan, decide, use tools, act on the environment, observe results and adapt - with limited human step-by-step guidance. A plain chatbot answers one prompt; an agent completes a multi-step task.

Core components of an AI agent

Component	Role
Brain (LLM)	Reasons, understands the goal and decides the next action.
Planning	Breaks a goal into sub-tasks; can re-plan when something fails (e.g. Chain-of-Thought, ReAct).
Memory	Short-term = current context window; long-term = vector database / stored notes.
Tools / Actions	APIs, web search, code execution, databases, file access, browsers. Often connected via standards like MCP.
Feedback loop	Observe result of each action, evaluate, then continue or correct.

The agent loop

Goal -> Plan -> Act (call a tool) -> Observe result -> Reflect -> Repeat until done. Example: "Find the top 3 papers on Swahili NLP and summarize them" - the agent searches, opens pages, extracts key points, checks quality and writes the summary.

Common patterns and frameworks

- **Patterns:** ReAct (reason + act), reflection/self-critique, tool use, planning, and **multi-agent systems** where specialized agents (researcher, coder, reviewer) collaborate.
- **Frameworks:** LangChain, LangGraph, LlamaIndex, CrewAI, AutoGen, and vendor agent SDKs.
- **Applications:** coding assistants, customer support, research and data analysis, workflow automation, personal assistants.

Risks and good practice

- **Hallucination and compounding errors** - mistakes can grow across many steps; add verification steps.
- **Safety and security** - limit permissions (least privilege), guard against prompt injection, sandbox code execution.
- **Human-in-the-loop** - require approval for high-impact actions (payments, deleting data, sending emails).
- **Cost, latency and evaluation** - log every step, set step limits, and test agents on realistic tasks.

5. Quick Comparison

	Machine Learning	Deep Learning	Agentic AI
Scope	Algorithms that learn from data	ML using multi-layer neural networks	Goal-driven, autonomous systems
Data needs	Small to medium	Large	Uses pre-trained models + tools
Feature work	Often manual	Learned automatically	Handled by the underlying models
Output	Prediction / classification	Prediction / generation	Completed multi-step tasks and actions
Example	Random Forest for churn	CNN for ASL sign images	Agent that researches and writes a report

Relationship: AI > Machine Learning > Deep Learning > LLMs > Agentic AI (agents are built on top of LLMs and other models by adding planning, memory and tools).